

Canopy and pruning management

Spatial scale: vineyard

About this sustainable management strategy

Canopy and pruning management aim to maintain a balanced vine structure that supports healthy grape production while improving the vineyard's resilience to changing climatic conditions. By regulating canopy density, shoot growth, and fruit exposure, growers can create a favourable microclimate around the grape clusters that reduces heat stress, improves air circulation, and promotes even ripening.

A combination of practices can be used throughout the growing season, including winter pruning, shoot positioning, shoot and cluster thinning, hedging, and selective leaf removal around the fruit zone. The choice and timing of each operation depend on the grape variety, vine vigour, local climate, and production objectives. In warmer regions, maintaining partial leaf cover can protect bunches from excessive solar radiation and sunburn, while cooler or more humid conditions may require a more open canopy to improve ventilation and reduce disease pressure.

Well-managed canopies improve light distribution, reduce excessive humidity within the vine, and help maintain a balance between vegetative growth and fruit development. As part of an integrated vineyard management approach, these practices can contribute to more stable yields, improved grape quality, and lower reliance on chemical plant protection.

Key ecosystem services:

- Microclimate regulation (*improved vineyard microclimate and reduced heat stress*)
- Natural pest control (*better air circulation and lower disease pressure*)
- Water regulation (*reduced plant water stress through balanced canopy development*)
- Improved grape quality and production stability
- Biodiversity enhancement (*support for beneficial organisms through reduced pesticide requirements*)

Considerations for successful implementation

Careful timing of operations

The timing of pruning, leaf removal, hedging, and thinning strongly influences vine growth, fruit development, and disease risk. Management decisions should respond to vine phenology and weather conditions rather than follow fixed calendar dates.

Balanced canopy structure

Removing too much foliage can expose grape clusters to excessive sunlight and high temperatures, increasing the risk of sunburn, dehydration, and quality losses. Maintaining sufficient leaf area is essential for photosynthesis and long-term vine health.

Labour planning and technical knowledge

Many canopy management operations require skilled labour and careful observation throughout the season. Selecting the most appropriate combination of practices can improve efficiency while achieving production and climate adaptation objectives.